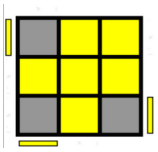
OLL Algorithms (Orientation of Last Layer)

Developed by Feliks Zemdegs
and Andy Klise

Algorithm Presentation Format

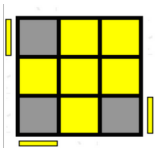


Suggested algorithm here
Alternative algorithms here
OLL Case Name - Probability = 1/x

Round brackets are used to segment algorithms to assist memorisation and group move triggers.

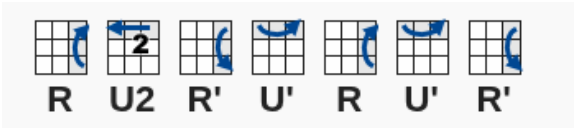
It is recommended to learn the algorithms in the order presented.

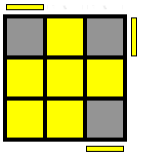
All Edges Oriented Correctly



R U2 R' U' R U' R'
y' R' U' R U' R' U2 R

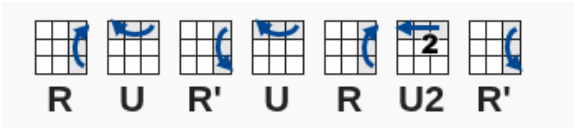
OCLL6 - 26 - Probability = 1/54

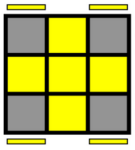




R U R' U R U2' R'
y' R' U2' R U R' U R

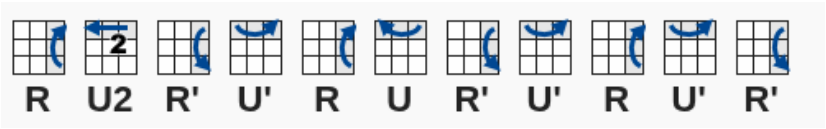
OCLL7 - 27 - Probability = 1/54

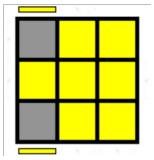




(R U2 R') (U' R U R') (U' R U' R')
y (R U R' U) (R U' R' U) (R U2' R')

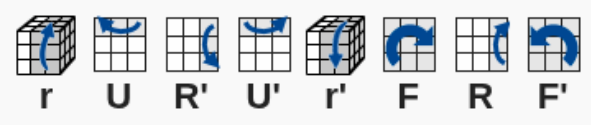
OCLL1 - 21 - Probability = 1/108



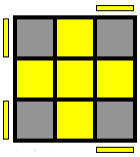


(r U R' U') (r' F R F')
y (R U R D) (R' U' R D') R2'

OCLL4 - 24 - Probability = 1/54

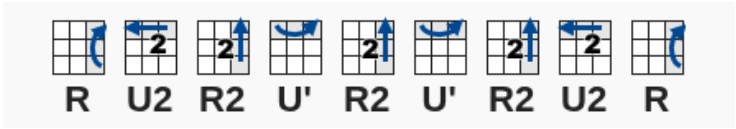


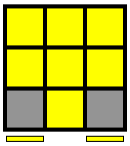
All Edges Oriented Correctly



R U2' R2' U' R2 U' R2' U2' R

OCLL2 - 22 - Probability = 1/54

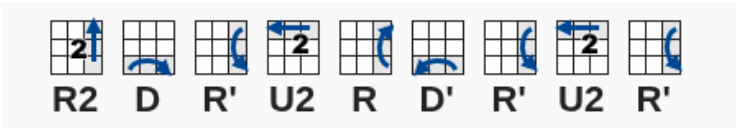


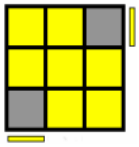


R2 D (R' U2 R) D' (R' U2 R')

y2 R2' D' (R U2 R') D (R U2 R)

OCLL3 - 23 - Probability = 1/54

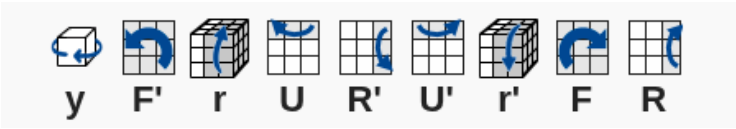




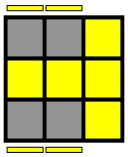
y F' (r U R' U') r' F R

x (R' U R) D' (R' U' R) D x'

OCLL5 - 25 - Probability = 1/54

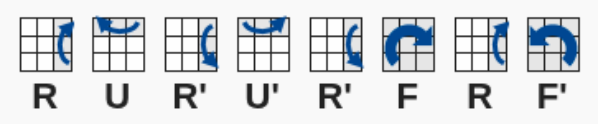


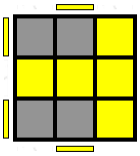
T-Shapes



(R U R' U') (R' F R F')

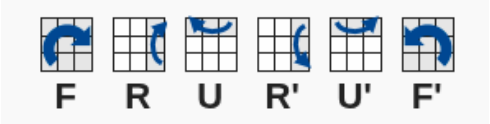
T1 - 33 - Probability = 1/54



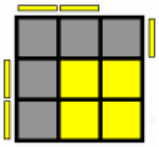


F (R U R' U') F'

T2 - 45 - Probability = 1/54

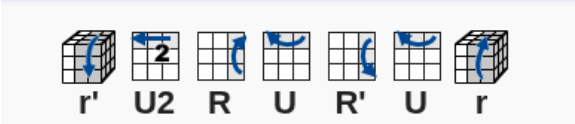


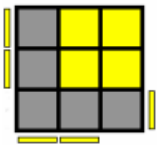
Squares



(r' U2' R U R' U r)

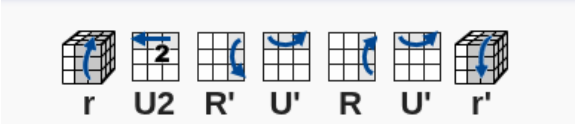
S1 - 5 - Probability = 1/54



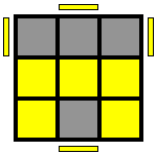


(r U2 R' U' R U' r')

S2 - 6 - Probability = 1/54



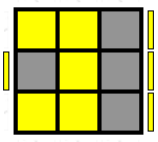
C-Shapes



(R U R2' U') (R' F R U) R U' F'

C1 - 34 - Probability = 1/54



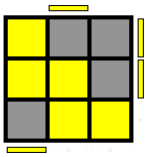


R' U' (R' F R F') U R

C2 - 46 - Probability = 1/54

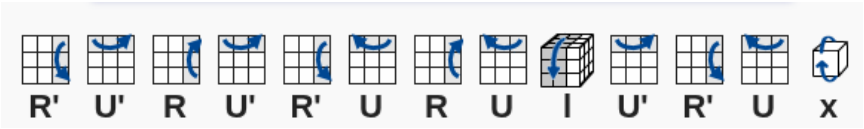


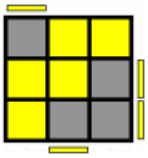
W-Shapes



(R' U' R U') (R' U R U) l U' R' U x
y2 (R U R' F') (R U R' U') (R' F R U') (R' F R F')

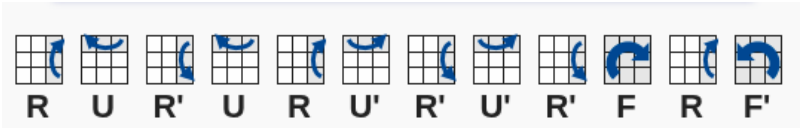
W1 - 36 - Probability = 1/54



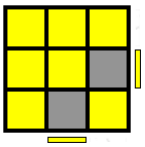


(R U R' U) (R U' R' U') (R' F R F')

W2 - 38 - Probability = 1/54

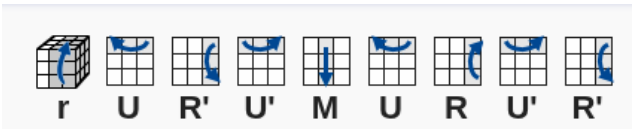


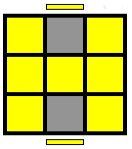
Corners Correct, Edges Flipped



(r U R' U') M (U R U' R')

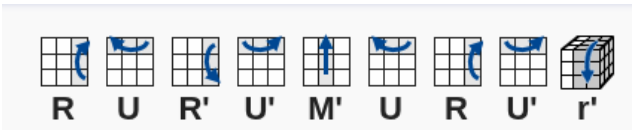
E1 - 28 - Probability = 1/54



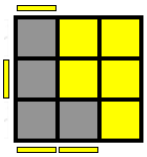


(R U R' U') M' (U R U' r')

E2 - 57 - Probability = 1/108

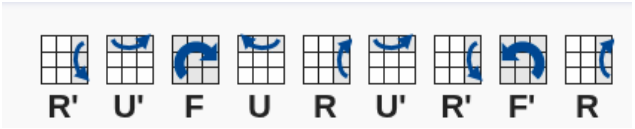


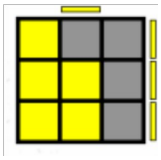
P-Shapes



(R' U' F) (U R U' R') F' R

P1 - 31 - Probability = 1/54



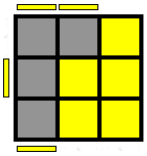


y R' U' F' U F R

f' (L' U' L U) f

P3 - 43 - Probability = 1/54

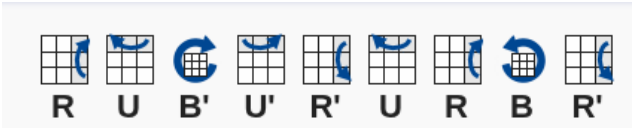


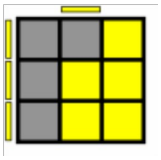


R U B' (U' R' U) (R B R')

S (R U R' U') (R' F R f')

P2 - 32 - Probability = 1/54

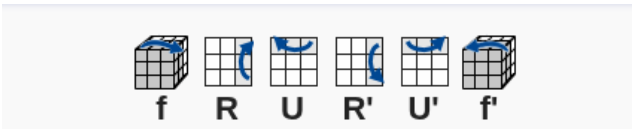




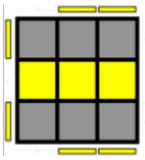
f (R U R' U') f'

y2 F (U R U' R') F'

P4 - 44 - Probability = 1/54

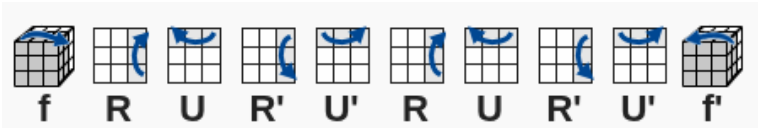


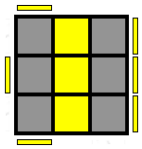
I-Shapes



f (R U R' U') (R U R' U') f'
y2 F (U R U' R') (U R U' R') F'

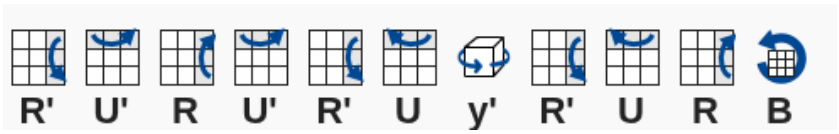
I1 - 51 - Probability = 1/54

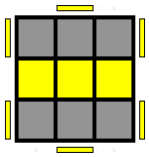




(R' U' R U' R' U) y' (R' U R) B
(R U R' U R U') y (R U' R') F'

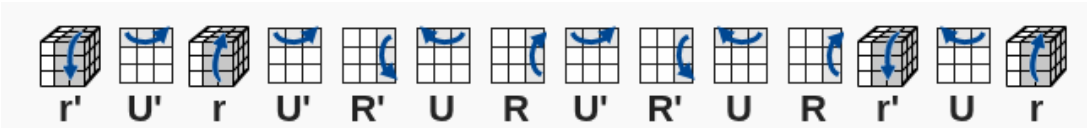
I2 - 52 - Probability = 1/54

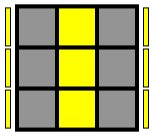




r' U' r (U' R' U R) (U' R' U R) r' U r

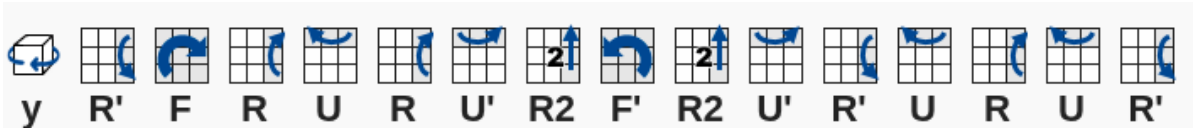
I4 - 56 - Probability = 1/108



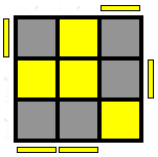


y (R' F R U) (R U' R2' F') R2 U' R' (U R U R')

I3 - 55 - Probability = 1/108



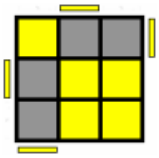
Fish Shapes



(R U R' U') R' F (R2 U R' U') F'
(R' U' R) y r U' r' U r U r'

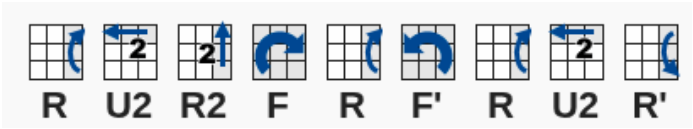
F1 - 9 - Probability = 1/54

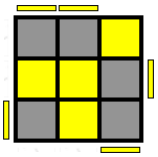




(R U2') (R2' F R F') (R U2' R')

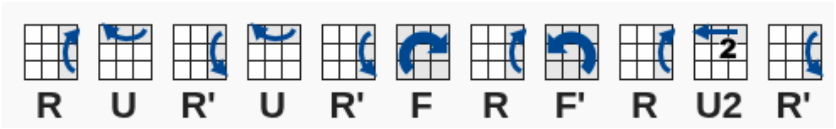
F3 - 35 - Probability = 1/54

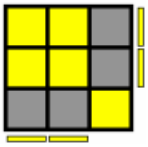




(R U R' U) (R' F R F') (R U2' R')
(R U R') y (R' F R U') (R' F' R)

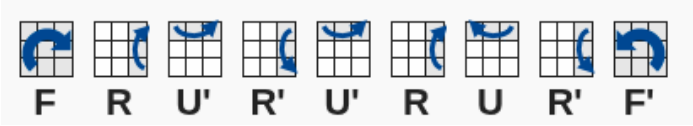
F2 - 10 - Probability = 1/54



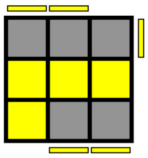


F (R U' R' U') (R U R' F')

F4 - 37 - Probability = 1/54

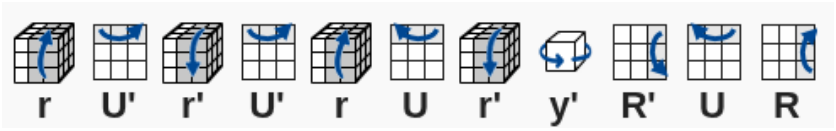


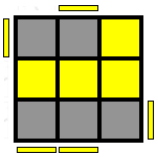
Knight Move Shapes



(r U' r') (U' r U r') y' (R' U R)
F U R U' R2' F' R U (R U' R')

K1 - 13 - Probability = 1/54

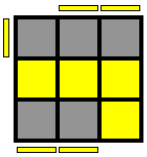




(r U r') (R U R' U') (r U' r')

K4 - 16 - Probability = 1/54

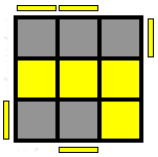




(R' F R) (U R' F' R) (F U' F')

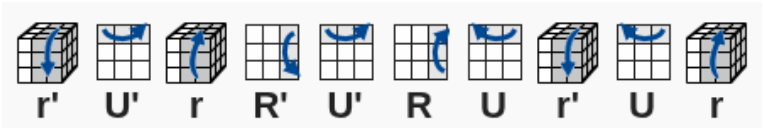
K2 - 14 - Probability = 1/54



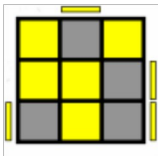


(r' U' r) (R' U' R U) (r' U r)

K3 - 15 - Probability = 1/54

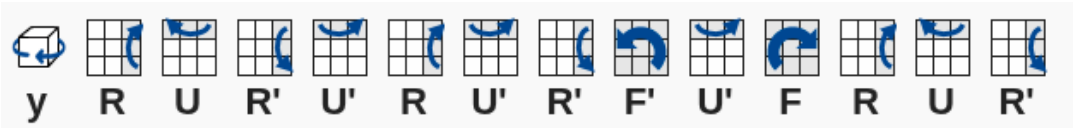


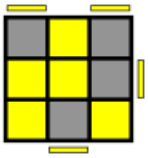
Awkward Shapes



y (R U R' U') (R U' R') (F' U' F) (R U R')
M U (R U R' U')(R' F R F') M'

A1 - 29 - Probability = 1/54

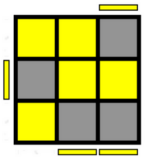




(R U R' U R U2' R') F (R U R' U') F'

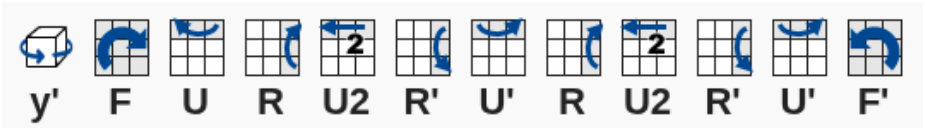
A3 - 41 - Probability = 1/54

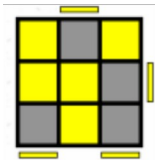




y' F U (R U2 R' U') (R U2 R' U') F'
y' (F R' F) (R2 U' R' U') (R U R') F2

A2 - 30 - Probability = 1/54



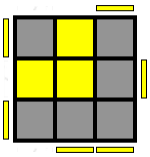


(R' U' R U' R' U2 R) F (R U R' U') F'
y (R' F R F') (R' F R F') (R U R' U') (R U R')

A4 - 42 - Probability = 1/54

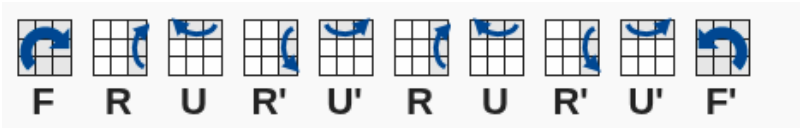


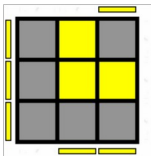
L-Shapes



F (R U R' U') (R U R' U') F'

L2 - 48 - Probability = 1/54

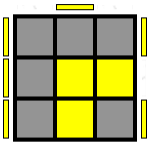




r U' r2' U r2 U r2' U' r

L3 - 49 - Probability = 1/54

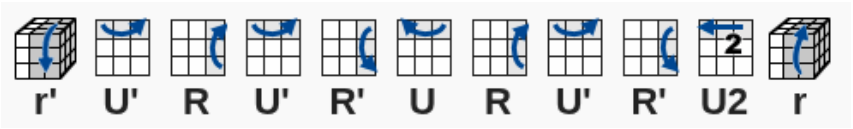


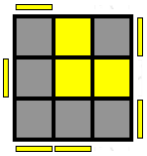


(r' U' R U') (R' U R U') R' U2 r

y r' U2' R (U R' U' R) (U R' U r)

L5 - 53 - Probability = 1/54

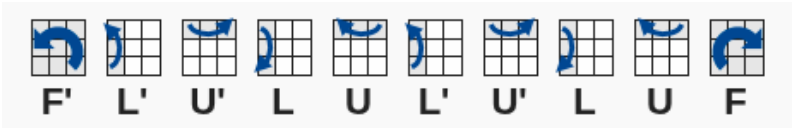


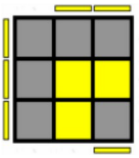


F' (L' U' L U) (L' U' L U) F

R' U' (R' F R F') (R' F R F') U R

L1 - 47 - Probability = 1/54

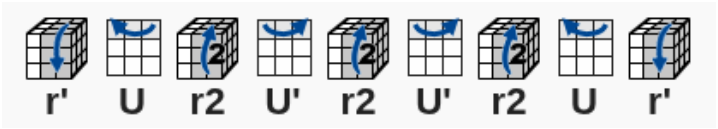


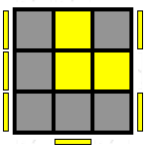


r' U r2 U' r2' U' r2 U r'

y' (R U2 R' U' R U' R') F (R U R' U') F'

L4 - 50 - Probability = 1/54

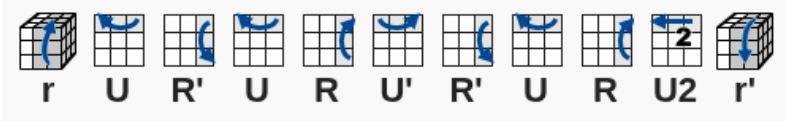




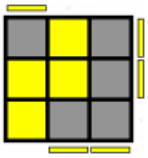
(r U R' U) (R U' R' U) R U2' r'

y' (r U2 R' U') (R U R' U') R U' r'

L6 - 54 - Probability = 1/54

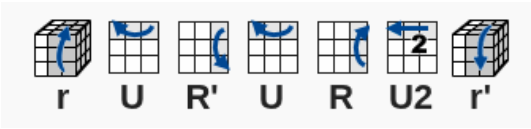


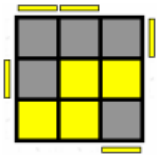
Lightning Bolts



(r U R' U R U2' r')

B1 - 7 - Probability = 1/54

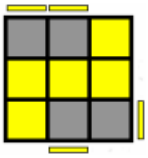




r' (R2 U R' U R U2 R') U M'

B3 - 11 - Probability = 1/54

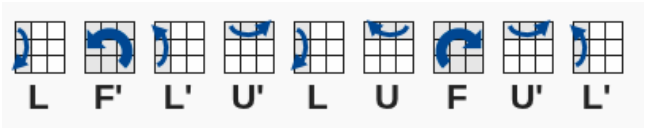


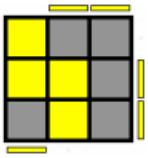


(L F') (L' U' L U) F U' L'

F (R U R' U') F' (R' U' R U' R' U2 R)

B5 - 39 - Probability = 1/54

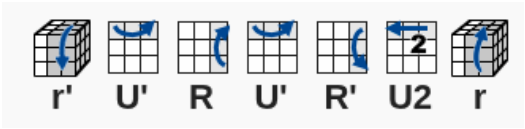


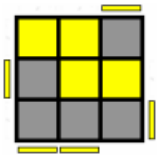


(r' U' R U' R' U2 r)

y2 l' U' L U' L' U2 l

B2 - 8 - Probability = 1/54

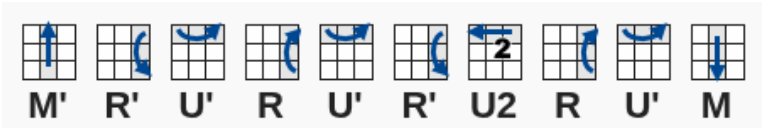


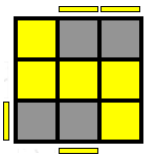


M' (R' U' R U' R' U2 R) U' M

y F (R U R' U') F' U F (R U R' U') F'

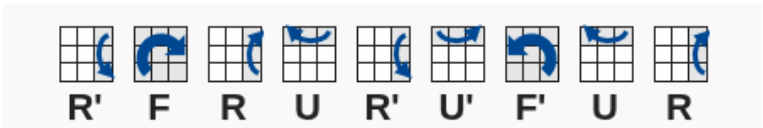
B4 - 12 - Probability = 1/54



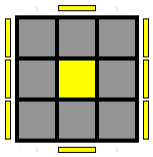


(R' F) (R U R' U') F' U R

B6 - 40 - Probability = 1/54



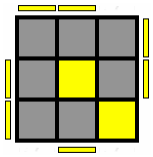
No Edges Flipped Correctly



(R U2') (R2' F R F') U2' (R' F R F')

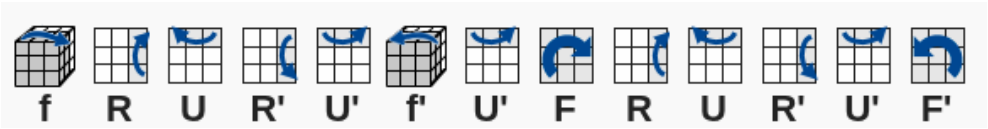
O1 - 1 - Probability = 1/108

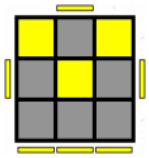




f (R U R' U') f' U' F (R U R' U') F'

O3 - 3 - Probability = 1/54



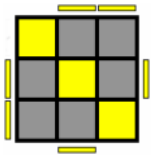


y R U2' (R2' F R F') U2' M' (U R U' r')

(r U R' U R U2 r') (r' U' R U' R' U2 r)

O6 - 18 - Probability = 1/54



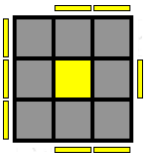


(R U R' U) (R' F R F') U2' (R' F R F')

O5 - 17 - Probability = 1/54



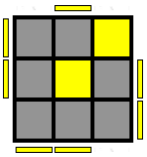
No Edges Flipped Correctly



F (R U R' U') F' f (R U R' U') f'
y (r U r') U2 R U2' R' U2 (r U' r')

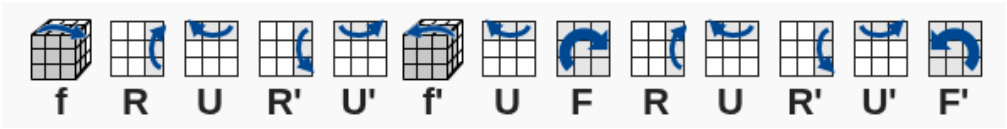
O2 - 2 - Probability = 1/54

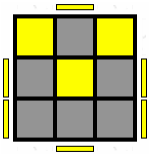




f (R U R' U') f' U F (R U R' U') F'

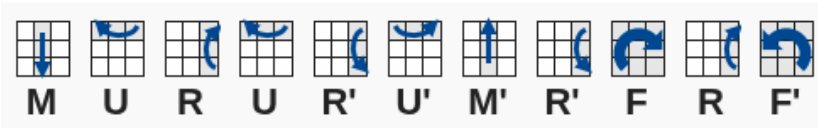
O4 - 4 - Probability = 1/54

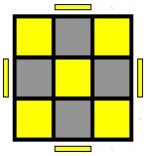




M U (R U R' U') M' (R' F R F')

O7 - 19 - Probability = 1/54





M U (R U R' U') M2' (U R U' r')
(r U R' U') M2' (U R U' R') U' M'

O8 - 20 - Probability = 1/216

